



United International University

School of Science and Engineering
Final Examination; Year 2025; Trimester: Summer
Course: PHY 2105; Title: Physics; Sec: All
Full Marks: 40, Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

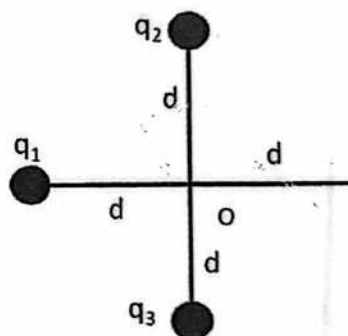
Questions no 1, 2, 3, 4 are mandatory to answer. Answer any one from question no 5 and 6.

1. (a) If two opposite charges (one positive and one negative) are placed on the same axis, is it possible to get a point on the axis between the charges where there is no electric field? Justify your answer. 2 CO1
- (b) Robert wanted to warm his sandwich and he used a microwave oven. Considering water molecules present in sandwich as dipoles, discuss how this oven can increase the temperature of the sandwich? 2 CO1
- (c) When does a dipole experiences no torque in an electric field? Explain briefly with diagram. 2 CO1

2. (a) John has pulled out some electron from a ball making a net charge of $+3.2 \times 10^{-15}$ C. How much electrons have been pulled out? 2 CO3

(b) Three charges lie on the x axis: $q_1 = +20 \mu\text{C}$ at the origin, $q_2 = -10 \mu\text{C}$ at $x = 4$ m, $q_3 = +40 \mu\text{C}$ at $x = 9$ m. Calculate (i) magnitude of net force on q_1 , and (ii) direction of net force on q_1 . 3 CO3

(c) Figure 01 shows below three particles with charge $q_1 = +Q$, $q_2 = +2Q$, and $q_3 = -8Q$, each a distance d from the origin. (i) What net electrostatic force is produced at the center point (origin, O) if there is a charge $q_0 = +1$ nC? Here, $Q = 5$ nC and $d = 4$ m. (ii) What is the direction of the net force at the origin? 3 CO3



3. (a) An electric dipole consists of charge $+5e$ and $-5e$ separated by 0.4 nm. It is in an electric field of strength 8×10^9 N/C. Calculate the magnitude of the torque on the dipole when the dipole moment is (i) parallel to and (ii) perpendicular to the electric field. 3 CO3

(b) Ammonium hydroxide (NH_4OH) is a molecule that has a permanent dipole moment is 5.98×10^{-30} Cm. 3 CO3

(i) What is the dipole distance of ammonia base molecule? The number of protons in N, O, and H are 7, 8, and 1, respectively.

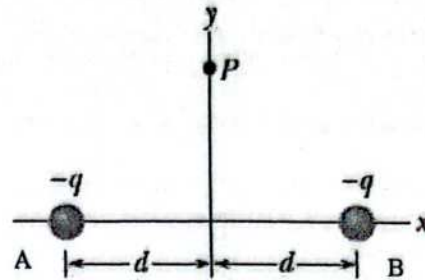
(ii) If the molecule with $\vec{P} = 5.98 \times 10^{-28} \hat{k} \text{ Cm}$ is placed in an electric field of $\vec{E} = 24 \hat{i} \text{ N/C}$, what is the potential energy of the dipole?

- (c) Four charges $+2q$, $+4q$, $-2q$ and $-3q$ are placed at the corners of a square. (i) Draw the arrangement of the charges, and (ii) Calculate the magnitude and direction of electric field at the intersection of the diagonals of the square of side 4 cm if $q = 2$ nC. 3 CO3

4. (a) Two equal charges of 2×10^{-3} C are shown in figure below. Each produces an electric field at point P (0,4) on Y axis. Consider the units in S. I. unit system. Here, $d = 1$ m

(i) What is the magnitude of the electric potential at point P?

(ii) If we replace the $-q$ charge at position A with a charge Q of unknown value such that the net potential at point P will be zero, what will be the value of Q? Calculate the value.



(b) Four charges $q_1 = +20$ nC, $q_2 = -30$ nC, $q_3 = +30$ nC and $q_4 = -20$ nC are situated at the circumference of a circle. If the radius of the circle is $r = 2$ m, what net electric potential at the center of the circle?

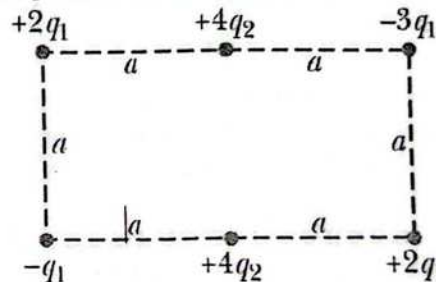
(c) The ammonia molecule NH_3 has a permanent electric dipole moment equal to $3D$, where $1D = 1$ Debye unit $= 3.4 \times 10^{-30}$ Cm. Calculate the electric potential due to an ammonia molecule at a point 60.0 nm away along the axis of the dipole. (Set $V = 0$ at infinity)

5. (a) Show that, the equation of electric field at a point due to a uniformly distributed infinite line charge or ring charge is

$$E = \frac{1}{4\pi\epsilon_0} \frac{qz}{(z^2 + R^2)^{3/2}}$$

(b) Show that, a potential energy U is associated with the orientation of the dipole moment in the field such that, $U = -\vec{P} \cdot \vec{E}$. Calculate the condition for maximum and zero potential energy. Here, $\vec{P}$ is the dipole moment and $\vec{E}$

6. (a) Figure below shows a rectangular array of charged particles fixed in place, with distance a and the charges shown as integer multiples of $q_1 = q$ and $q_2 = 2q$. With $V = 0$ at infinity, calculate the net electric potential at the rectangle's center.



(b) Show that, at a distance r from an electric dipole with dipole moment magnitude $p = qd$, the electric potential of the dipole is,

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

Here, the other symbols have their usual meanings.

CO1: Define different physical quantities with examples. **CO2:** Derive/Show the various equations of electric field, electric potential, electric dipole, dipole moment, electrostatic force, etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of electric charge, electric field, electric potential, electric dipole moment, and electrostatic force, etc.